

Homework 3 - Time series

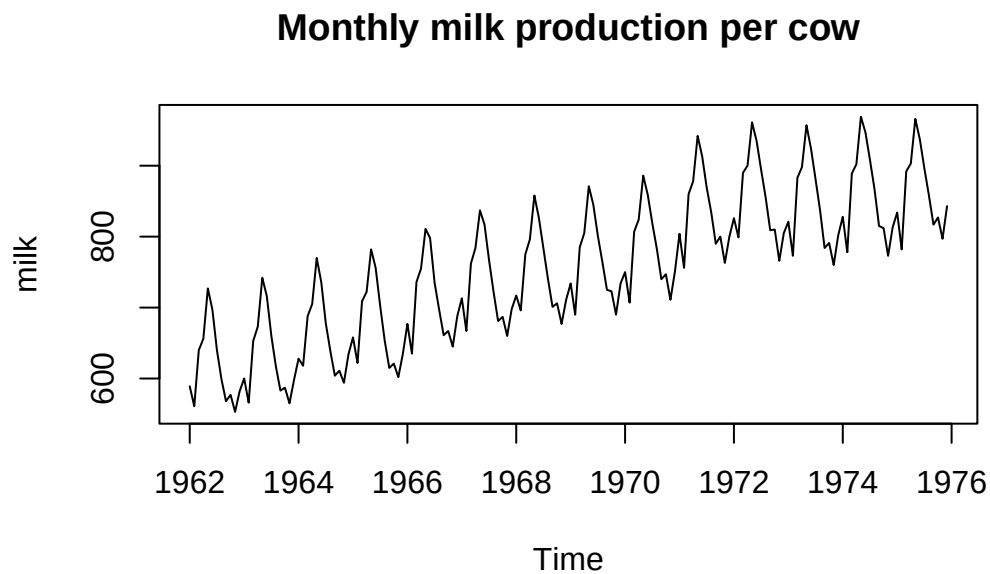
Mateus Auza Cruz

2026-03-28

```
milk=read.table("milk.txt", header=F)
milk= ts(milk, start=1962, frequency=12)
```

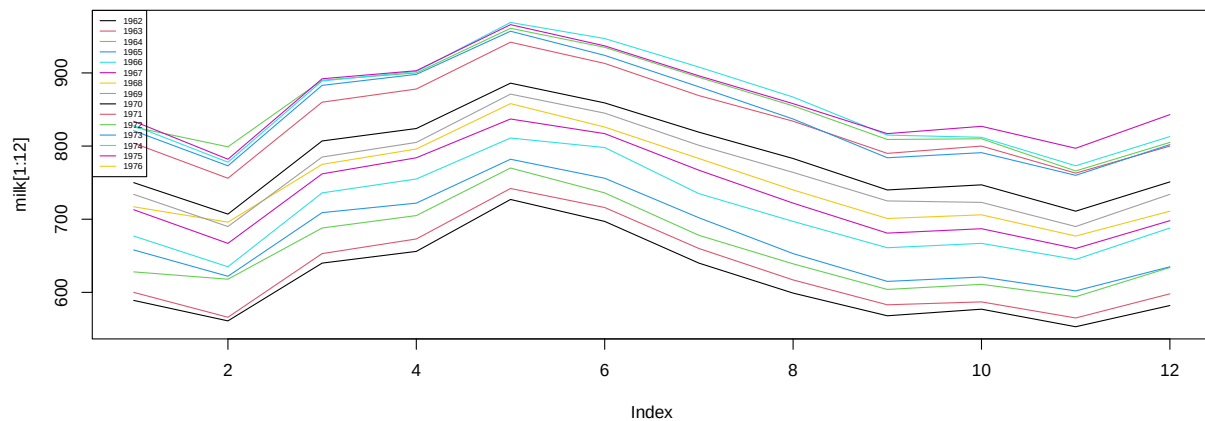
Our time series analysis looks like this:

```
plot(milk, ylab="milk", main="Monthly milk production per cow")
```



From this plot we can see that the data is non stationary and there seems to have a seasonal and an non stable variance. To better visualize the seasonal component (and to check that indeed we have an $S = 12$) we will do the following plot:

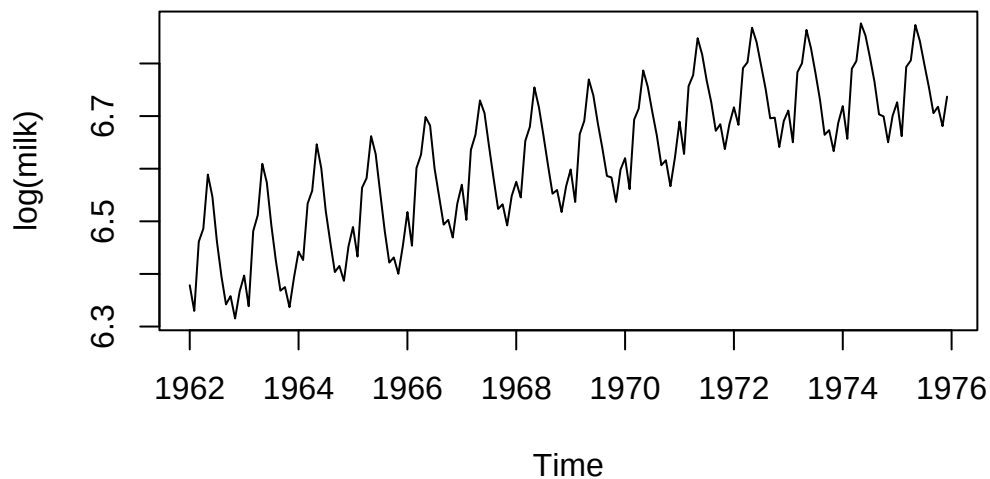
```
plot(milk[1:12], type="l", ylim=c(min(milk), max(milk)))
for(i in 2:16){
  lines(milk[12*(i-1)+(1:12)],col=i)
}
legend("topleft", legend = 1962:1976, lty = 1, col = 1:15, cex = 0.5)
```



The plot suggests a trend and seasonality, so we consider $d = 1$ and $D = 1$. We also have a non stationary variance. Thus we need to correct this. First we will correct the variance using a log transformation.

```
log.milk=log(milk)
plot(log.milk, ylab="log(milk)", main="Monthly milk production per cow (log scale)")
```

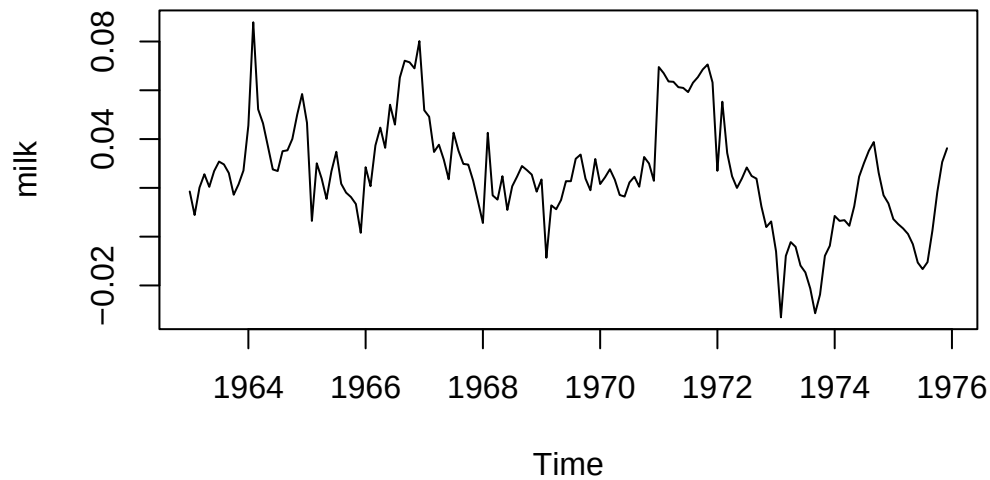
Monthly milk production per cow (log scale)



Then we remove the seasonality:

```
no.season=diff(log.milk,lag=12)
plot(no.season, main = "Seasonality in 'milk_log' (S = 12)", ylab="milk")
```

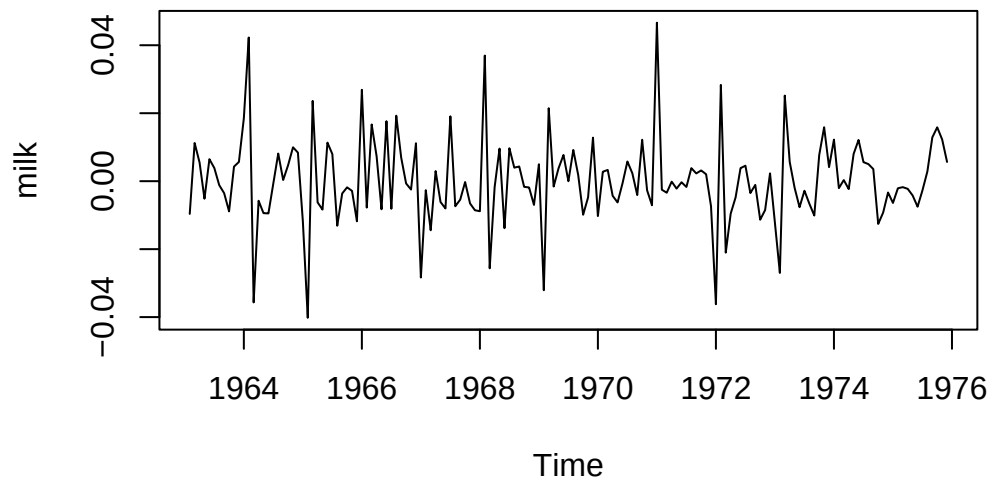
Seasonality in 'milk_log' (S = 12)



Now we remove the linear trend by differencing once.

```
no.dep=diff(no.season, lag=1)
plot(no.dep,ylab= "milk", main= "Remove linear trend in 'milk_noseason' (d = 1)")
```

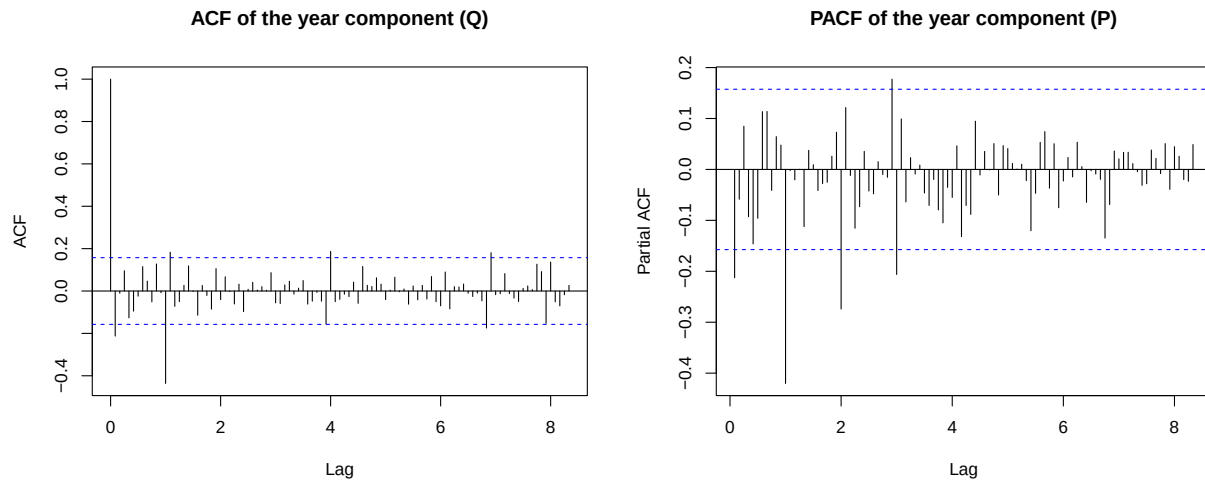
Remove linear trend in 'milk_noseason' (d = 1)



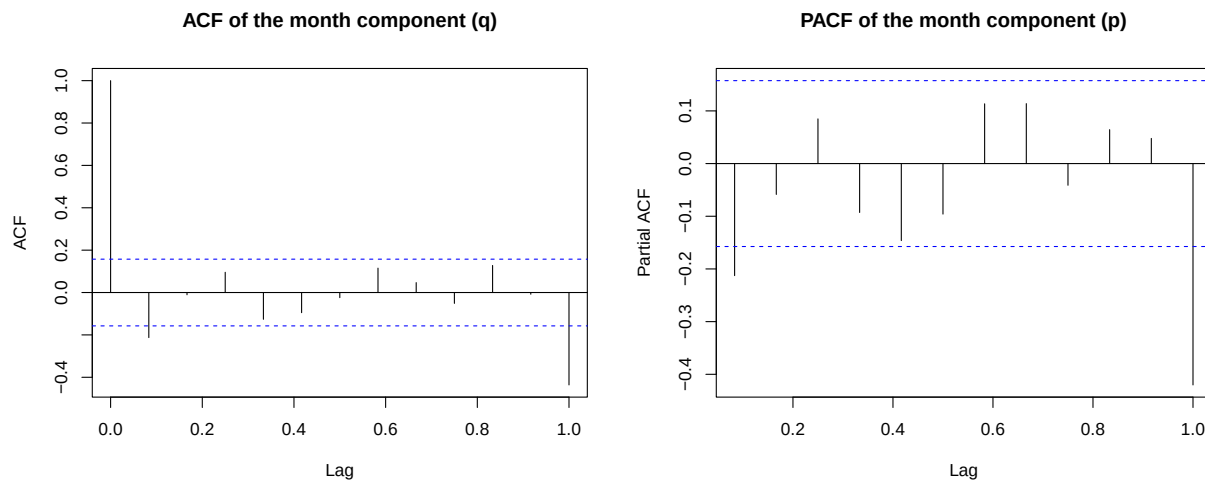
Now, we can see that this time series looks **stationary** and thus we can use it to forecast data.

Now our goal is to fit the best model for our data, this will be made through the stationary data. To do that, we will plot the **acf** and the **pacf**.

```
# Yearly component
par(mfrow=c(1,2))
acf(no.dep, lag.max=100, main="ACF of the year component (Q)")
pacf(no.dep, lag.max=100, main="PACF of the year component (P)")
```



```
# Monthly component
par(mfrow=c(1,2))
acf(no.dep, lag.max=12, main="ACF of the month component (q)")
pacf(no.dep, lag.max=12, main="PACF of the month component (p)")
```



From the ACF and PACF plots, we can identify possible components for a SARIMA model.

First, looking at the **ACF of the seasonal (yearly) component**, we see a clear spike at lag 12 that goes beyond the 95% confidence bounds. After that, there is only a small spike around lag 48, and the values slowly decrease. This suggests a **seasonal MA component**, with order ($Q = 0$) or ($Q = 1$), since the pattern seems to tail off.

Next, in the **PACF of the seasonal component**, we notice significant spikes at multiples of 12 (lags 12, 24, and 36), although they become weaker after lag 36. This pattern is typical of a **seasonal AR component**, so we can reasonably choose ($P = 1$).

For the **ACF of the non-seasonal (monthly) component**, there is a spike at lag 1 (besides lag 0), followed by a gradual decrease. This points to a **non-seasonal MA component** with order ($q = 0$) or ($q = 1$).

Finally, the **PACF of the non-seasonal component** shows a clear spike at lag 1 and then quickly decreases. This suggests a **non-seasonal AR component** with order ($p = 0$) or ($p = 1$).

Based on these observations, we consider the following candidate models:

- SARIMA(1, 1, 0)(1, 1, 0)
- SARIMA(0, 1, 1)(1, 1, 0)
- SARIMA(1, 1, 1)(1, 1, 0)
- SARIMA(1, 1, 0)(1, 1, 1)
- SARIMA(0, 1, 1)(1, 1, 1)
- SARIMA(1, 1, 1)(1, 1, 1)

We will now try to choose the best model through the AIC/BIC criteria:

```
fit1= arima(milk, order= c(1,1,0), seasonal= list(order=c(1,1,0), period=12))
fit2= arima(milk, order= c(0,1,1), seasonal= list(order=c(1,1,0), period=12))
fit3= arima(milk, order= c(1,1,1), seasonal= list(order=c(1,1,0), period=12))
fit4= arima(milk, order= c(1,1,0), seasonal= list(order=c(1,1,1), period=12))
fit5= arima(milk, order= c(0,1,1), seasonal= list(order=c(1,1,1), period=12))
fit6= arima(milk, order= c(1,1,1), seasonal= list(order=c(1,1,1), period=12))

data.frame(
  Model = paste0("fit", 1:6),
  AIC = c(AIC(fit1), AIC(fit2), AIC(fit3), AIC(fit4), AIC(fit5), AIC(fit6)),
  BIC = c(BIC(fit1), BIC(fit2), BIC(fit3), BIC(fit4), BIC(fit5), BIC(fit6))
)
```

Model	AIC	BIC
fit1	1081.589	1090.719
fit2	1082.128	1091.258
fit3	1083.529	1095.702
fit4	1067.918	1080.091
fit5	1068.034	1080.208
fit6	1069.780	1084.997

The three best models according to these criteria are: **SARIMA(1,1,0)(1,1,1)**, **SARIMA(0,1,1)(1,1,1)**, and **SARIMA(1,1,1)(1,1,1)**.

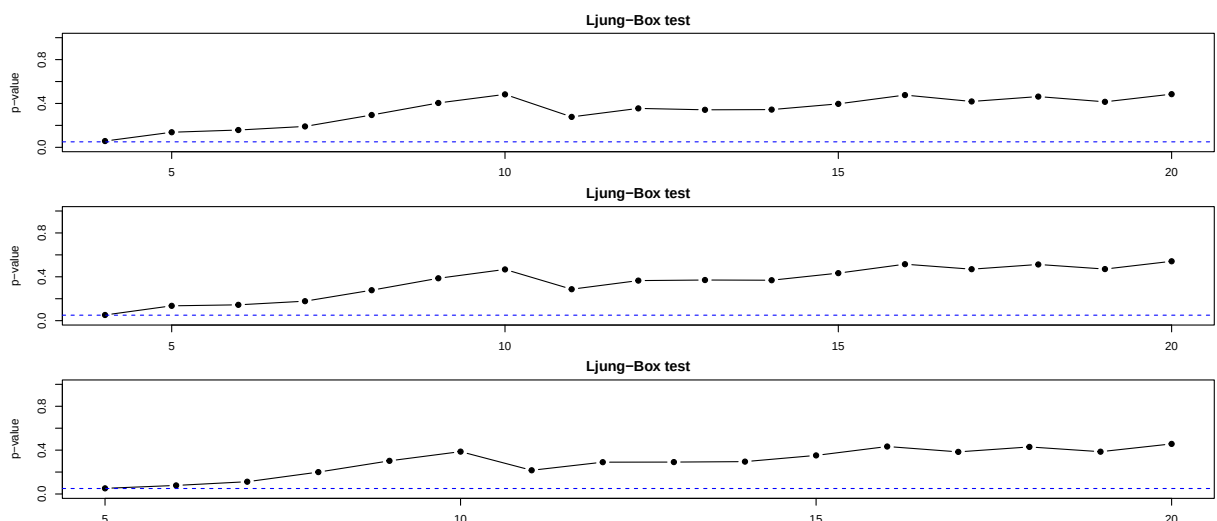
We now assess the adequacy of these models using the Ljung–Box test, in order to check for remaining autocorrelation in the residuals.

```
LjungBoxPlot <- function(res, ARord, MAord, lag.max = 20){
  pval <- numeric(lag.max)
  tot <- ARord + MAord
  for(k in (tot + 1):lag.max){
    pval[k] <- Box.test(res, lag = k, type = "Ljung", fitdf = tot)$p.value
  }

  plot((tot+1):lag.max, pval[(tot+1):lag.max],
       type="b", pch=19, ylim=c(0,1),
       xlab="Lag", ylab="p-value", main="Ljung-Box test")
  abline(h = 0.05, col="blue", lty=2)
}

resid4= residuals(fit4)
resid5= residuals(fit5)
resid6= residuals(fit6)

par(mfrow = c(3,1), mar = c(2, 6, 2, 1))
LjungBoxPlot(resid4, ARord = 2, MAord = 1)
LjungBoxPlot(resid5, ARord = 1, MAord = 2)
LjungBoxPlot(resid6, ARord = 2, MAord = 2)
```



The Ljung–Box test does not indicate significant autocorrelation in the residuals, suggesting that all three models provide an adequate fit. Among these models, **SARIMA(1,1,0)(1,1,1)** (fit4) has the lowest AIC, and is therefore selected as the final model.

Let us now estimate the parameters:

```
fit4
```

Call:

```
arima(x = milk, order = c(1, 1, 0), seasonal = list(order = c(1, 1, 1), period = 12))
```

Coefficients:

	ar1	sar1	sma1
	-0.2276	-0.0650	-0.5822
s.e.	0.0783	0.1188	0.0957

sigma^2 estimated as 52.55: log likelihood = -529.96, aic = 1067.92

The model can be written in backshift notation as

$$(1 - 0.2276B)(1 - 0.0650B^{12})(1 - B)(1 - B^{12})X_t = (1 - 0.5822B^{12})\varepsilon_t.$$

Expanding this expression:

$$\begin{aligned}X_t = & 1.2276 X_{t-1} - 0.2276 X_{t-2} \\& + 1.0650 X_{t-12} - 1.307394 X_{t-13} + 0.242394 X_{t-14} \\& - 0.0650 X_{t-24} + 0.079794 X_{t-25} - 0.014794 X_{t-26} \\& + \varepsilon_t - 0.5822 \varepsilon_{t-12}.\end{aligned}$$

Now, that we have the equation of the model, we will now try to predict the 15 values ahead:

```
pred.ts <- predict(fit4, n.ahead = 15)

ts.tot <- ts(c(milk, pred.ts$pred), start = 1962, frequency = 12)

plot(ts.tot, type = "l", col = "blue",
     main = "Forecast of the milk time series")

lines(ts(pred.ts$pred + 1.96 * pred.ts$se,
        start = tsp(milk)[2] + 1/12, frequency = 12),
      lty = 2, col = "red")

lines(ts(pred.ts$pred - 1.96 * pred.ts$se,
        start = tsp(milk)[2] + 1/12, frequency = 12),
      lty = 2, col = "red")
```

